Customer Churn Analysis Report

Project Name: Customer Churn Analysis using Python & SQLite

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Tools Used: Python, Pandas, NumPy, SQLite, Matplotlib, Seaborn, Jupyter Notebook

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Project Objective

The objective of this project is to analyze customer churn behavior using customer, subscription, and support datasets stored in a SQLite database. The analysis identifies factors responsible for churn and provides insights that can help businesses improve customer retention.

Dataset Overview

The project uses three tables stored inside **customer_churn.db**.

1. db_customer

Contains customer demographic information.

Columns include:

- customerid
- name
- country
- state
- gender
- dob
- interests
- pincode

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2. db_subscription

Contains subscription information.

Columns include

- customerid
- subscription_start_date
- subscription_type
- renewal_date
- plan_type
- contract_type
- cancellation_date
- cancellation_reason
- monthly_charges
- cltv
- churn_score

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3. db_support

Contains customer support information.

Columns include

- customerid
- complaint_date
- escalations
- csat_score
- comment

Data Preparation

The following preprocessing steps were performed.

- Connected SQLite database using sqlite3
- Loaded all tables into Pandas DataFrames

- Inspected schema using PRAGMA table_info
- Merged tables using customerid
- Converted date columns into datetime format
- Created new analytical columns
- Generated churn_flag
- Created churn_risk categories
- Prepared encoded dataset for correlation analysis

Data Merging

All three tables were merged using Left Join.

Subscription

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Customer

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Support

Final merged dataset contains customer demographics, subscription history and support information.

Exploratory Data Analysis

The following visualizations were created.

Monthly Churn Trend

Shows the number of customers cancelling every month.

Observation

- Churn increased during September.
- Other months experienced comparatively fewer cancellations.

Plan Type vs Churn

Purpose

Compare churn rate across different subscription plans.

Finding

Some plans experience higher customer churn than others.

Correlation Heatmap

Correlation was calculated after encoding categorical columns.

Important observations

- Churn Score has a strong positive correlation with Churn Flag.
- Escalations positively correlate with Churn Score.
- Churn Risk has a strong negative correlation with Churn Flag.
- Contract Type also influences churn behavior.

Pair Plot

Pairwise relationships between important variables.

Variables

- Plan Type
- Contract Type
- Churn Score
- Churn Flag
- Churn Risk
- Escalations

Purpose

Identify hidden relationships and data distribution.

Key Findings

Customer Behavior

Customers with higher churn scores are much more likely to leave.

Escalations

Customers having multiple escalations generally have higher churn probability.

Contract Type

Contract type influences customer retention.

Longer commitment contracts generally retain customers better than flexible plans.

Churn Risk

Customers categorized under high-risk segments require immediate attention.

Business Recommendations

Improve Customer Support

Reduce escalations by resolving complaints faster.

Early Warning System

Monitor customers having high churn scores and proactively contact them.

Retention Campaigns

Offer discounts and loyalty rewards before subscription cancellation.

Improve Subscription Plans

Review plans with higher churn and redesign pricing or benefits.

Monitor Customer Satisfaction

Improve CSAT scores through better customer service.

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- Matplotlib
- Seaborn
- Jupyter Notebook

Skills Demonstrated

- SQL Database Analysis
- SQLite
- Data Cleaning
- Data Merging
- Exploratory Data Analysis
- Data Visualization
- Correlation Analysis
- Customer Analytics
- Business Insight Generation
- Python Programming
- Pandas
- Seaborn

- Matplotlib

Conclusion

This project successfully analyzes customer churn by integrating customer, subscription, and support datasets. The analysis highlights key factors such as churn score, contract type, escalations, and churn risk that significantly influence customer retention. These insights can help organizations design targeted retention strategies, improve customer satisfaction, and reduce future churn.